

Skyscraper urbanisation: Visualising vertical growth, economic cycles and 3-D urbanisation

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Abstract

Skyscrapers, as symbols of vertical urbanisation, embody both material and symbolic dimensions of city growth. They have been associated with economic cycles, spatial constraints and global competition, yet their potential as comparative urbanisation metrics remains underexplored. Drawing on a global dataset of more than 10,000 skyscrapers taller than 100 metres, this study visualises their temporal evolution across nations and cities and introduces a relative 3-D urbanisation index to compare vertical and horizontal expansion. The results show that record-breaking towers often coincide with construction booms and sometimes precede downturns, lending partial support to the ‘Skyscraper Curse’. However, the correlation is uneven: state-led economies such as China and the UAE maintain steady construction momentum, while market-driven contexts such as the United States and Germany are more sensitive to global cycles. Within nations, similar contrasts emerge, with Shanghai’s volatility differing from Guangdong’s consistency. By linking skyscraper construction with built-up area growth, the 3-D index highlights marked disparities: only a handful of land-constrained states exceed an index value of 1, while most countries cluster near 0.7. Two trajectories emerge: constraint-driven verticalisation in small or dense territories, and expansion-driven horizontalism in larger states with extensive sprawl. Skyscrapers are thus better interpreted as temporal markers of ambition and governance rather than deterministic predictors of collapse, providing new insights into the uneven intersections of vertical and horizontal urbanisation.

Keywords

skyscraper, urbanisation, economic recession, competition, urban growth

Introduction

Urbanisation has long been conceptualised in two dimensions: horizontal expansion through land consumption and vertical growth through building height (Batty, 2018; Nethercote, 2018). Skyscrapers, in particular, represent a unique phenomenon: once completed, they no longer evolve individually but instead contribute to a broader system where the number and height of buildings transform in tandem with urbanisation processes (Batty, 2010). As such, skyscrapers serve both as a metric of short-term economic transformation and as a lens for understanding the formation of vertical urban morphology.

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Yet skyscrapers embody more than physical form. They represent political aspirations, economic ambition and global competition for prestige. Scholarship on vertical urbanism highlights that towers are embedded in wider power relations and governance structures. Graham and Hewitt (2012) emphasise the politics of verticality, noting how towers manifest authority, security and urban control. Olds (2014) links skyscraper megaprojects in Pacific Rim cities to global capital flows and intercity rivalry, while Sklair (2017) frames them as icons of capitalist globalisation, designed as much for symbolic visibility as for functionality. McNeill (2005) similarly argues that skyscrapers contribute to global city identity, concentrated in financial centres competing for status. Related scholarship has further developed the idea of a ‘vertical city’, extending urban theory to explicitly include the three-dimensional built environment (Appert & Montes, 2015; Charney, 2007).

Skyscrapers also intersect with moments of crisis. Kaika (2010) shows how architecture in London was reinvented during economic turbulence, as skylines were re-imagined to project resilience and renewal. This resonates with the ‘Skyscraper Curse’ (Thornton, 2018), which associates record-breaking towers with impending downturns. Recent economic analyses, however, caution against overstating this relationship, noting that correlations are inconsistent and shaped by planning lags and strategic interventions (Barr & Luo, 2017; O’Meara, 2019). Willis (1995) and Sudjic (2005) also emphasise the symbolic and political economy dimensions of skylines, showing how financial speculation, prestige and elite ambition often drive tall building projects.

Comparative work has also highlighted how skyscraper development reflects spatial constraints and governance contexts. In land-scarce cities such as Singapore or Hong Kong, verticalisation substitutes for limited horizontal expansion, whereas in market-driven contexts like the United States, skyscrapers reflect cycles of speculative real estate markets (Jenks & Tallon, 2008; Shatkin, 2014). Broader frameworks for analysing urban morphology and form similarly stress the need to integrate vertical and horizontal growth (Abel, 2003; Cuthbert, 2011).

Despite this growing literature, most research on skyscrapers remains descriptive or case-specific, often limited to emblematic cities or iconic towers. Comparative, quantitative analyses of vertical urbanisation across regions are rare, and the relationship between skyscraper growth, economic cycles and spatial constraints remains under-theorised. Moreover, horizontal and vertical urbanisation are often examined in isolation, rather than in their interdependence.

This paper addresses these gaps by focusing on three research questions:

- How do skyscraper construction patterns align with global and regional urbanisation trajectories?
- To what extent are record-breaking skyscrapers co-present with economic cycles, and what alternative explanations exist?
- How can vertical and horizontal urbanisation be systematically compared through visualisation?

By addressing these questions, this paper employs visualisation to contribute to the growing recognition of the need to study cities in three dimensions (Harris, 2015). The focus is not only on individual landmark tall buildings, but also on how cumulative vertical expansion interacts with horizontal urban growth and reflects broader urban dynamics. The following graphics offer a comparative lens through which these processes can be visualised.

Data and method

We compiled a spatiotemporally referenced dataset of 10,357 skyscrapers ≥ 100 m, sourced from SkyscraperPage and Kaggle – two open-access repositories documenting skyscrapers worldwide. The ≥ 100 m threshold aligns with the Council on Tall Buildings and Urban Habitat (CTBUH) definition, ensuring global comparability across time and space. After removing duplicate entries and geocoding each

building, the processed dataset comprises over 10,357 skyscrapers, distributed across 308 metropolitans/regions, 56 countries and eight continents. Countries with fewer than five skyscrapers were excluded to avoid statistical distortion. Built-up area sizes were derived from global urban footprint datasets, capturing functional urban areas rather than simply administrative boundaries.

To address the research questions, we visualised this dataset through two illustrations. The first depicts the evolution of skyscraper heights and numbers over time, analysed across various administrative levels, including regions, nations and continents and overlapped with economic cycles. The second explores the 3-D urbanisation by visualising the interrelationship between the volume of developed skyscrapers and the size of built-up areas, shedding light on the three-dimensional urbanisation patterns observed across different cities.

Visualising skyscraper construction as vertical growth with economic cycles

We first visualised the development of skyscraper construction across time and geography to understand how this form of vertical urbanisation has unfolded at multiple scales. Our analysis focuses on three geographic levels: continents, nations and regions. Nations and their regions were ranked by total skyscraper completions to date, with the top 50 selected as representative cases.

For each case, we mapped the historical height record as a line chart, with individual skyscrapers represented as dots positioned by year of completion and height. The density of dots, further visualised as heatmaps, reveals temporal clustering of skyscraper construction activity. Interconnecting curves trace the tallest buildings completed each year, thereby illustrating the progression of annual height records. Together, these two dimensions – the density of construction and the breaking of height records – capture the vertical growth of urbanisation.

This procedure was applied separately at the national and regional levels, after which the two scales were linked by mapping their administrative affiliations through a Sankey diagram. Nations and regions were colour-coded by continental affiliation. To illustrate the shifting leadership in global skyscraper construction, we included subfigures in which the width represents the number of skyscrapers completed each year. To capture inherent variation across hidden groups, we applied spatiotemporal clustering of all cities to delineate groupings, thereby reflecting how vertical growth unfolds both within and between groups. Finally, to explore the co-occurrence of skyscraper record-breaking and periods of economic decline, we overlaid the charts with well-known economic cycles.

Visualising 3-D urbanisation by comparing skyscraper construction with built-up areas

Cities differ in size, leading to varying modes of skyscraper development shaped by their urban landscapes. To compare vertical and horizontal expansion, we propose a relative 3-D urbanisation index (I) defined as:

$$I = \ln\left(\frac{H}{100} + 1\right) / \ln\left(\frac{S}{100} + 1\right)$$

Here, H denotes the cumulative height of all skyscrapers built within a given area, and S represents the cumulative size of the built-up area where skyscrapers are built. An index value of 1 corresponds to one skyscraper taller than 100 m per 100 km² of urban area. To normalise the cumulative skyscraper height, we divide by 100 – based on experimental adjustments – so that the resulting visualisation can be plotted on a well-organised canvas.

The 3-D urbanisation index captures national and regional variations in the interrelationship between vertical and horizontal development and is utilised to rank nations and regions accordingly. To further reveal

details not fully reflected by the index, we employed an overlapped proportional area chart to illustrate diverse modes of 3-D urbanisation across different administrative areas. We documented this in our second visualisation result.

Visualisation results

Skyscraper evolution and height records

Figure 1 depicts the construction timeline of skyscrapers. It shows a general co-evolution between the growing number of skyscrapers and their increasing heights across most nations and regions. This suggests that regions seeking to build more skyscrapers also tend to pursue greater heights. Importantly, record-breaking heights do not follow a linear trajectory; rather, they typically emerge after construction booms, which in turn elevate average skyscraper heights. This pattern indicates that reaching new height records often requires periods of economic expansion and the accumulation of capital to support such ambitious projects. Bursts of activity are particularly evident in China during the 2000s and 2010s, whereas the United States demonstrates steadier long-term growth. Middle Eastern nations, by contrast, display sudden but isolated surges associated with prestige-driven projects.

We ranked nations and their subnational regions based on the total number of skyscrapers, visualising the top 50 in Figure 1. The top 50 nations account for 7,065 skyscrapers, representing 98.4% of the global total with the completion time, while the top 50 subnational regions contribute 5,418 skyscrapers, or 75.5%. Nations with high rankings often include subnational regions or cities that are similarly ranked, such as China, the United States and Singapore. However, countries like Canada, Germany and Australia are represented by specific subnational regions or metropolitans that rank higher in skyscraper counts than their national standings.

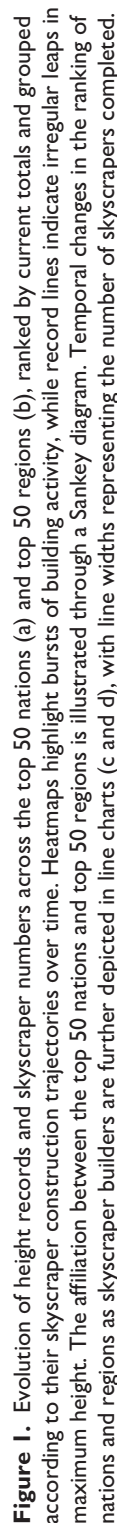
The top 50 nations are categorised into four groups:

- Skyline Pioneers: led by China and the United States, these countries boast the most and tallest skyscrapers across all periods;
- Skyline Marathoners: comprising mainly Western developed nations, these countries began building skyscrapers early but have pursued restrained development;
- Newcomers: countries such as those in South Asia, which started constructing skyscrapers post-21st century, though with relatively moderate heights;
- Bloomers: nations in the Middle East and similar regions that began building skyscrapers more recently but have focused on taller structures than the Newcomers.

For regional rankings, only three groups emerge, with no distinct Bloomers category due to more consistent height variations. China dominates across all groups, providing a majority of subnational regions, while US cities are primarily classified as Pioneers and Marathoners. Additionally, China contributes significantly to the Newcomers category, alongside cities from nations categorised as Newcomers. Throughout the rankings, the United States, China, Canada and Australia have maintained their positions in the top 10 since their entry. Notably, Tokyo and New York continue to lead all cities within the top 10 club.

Skyscrapers and economic cycles

By aligning periods of economic downturn with skyscraper construction patterns, we observed that the emergence of record-breaking towers often coincided with dramatic growth spurts followed by subsequent recessions. This phenomenon lends support to the validity of the widely discussed ‘Skyscraper Curse’, as encapsulated in the Skyscraper Index theory, which posits that the world’s tallest buildings frequently appear



on the eve of major economic crises (Thornton, 2018). While this trend is broadly consistent, significant national deviations can be observed.

In the United States, extreme skyscraper heights have historically foreshadowed severe economic crises, including the Panic of 1907, the Great Depression, the era of Stagflation and the Financial Crisis of 2008. By contrast, in China, height records have aligned more closely with events such as the Asian Financial Crisis, the Dot-com Bubble and the COVID-19 pandemic. At the regional scale, large cities reveal varying degrees of correspondence with economic downturns. In the US, for example, record heights in New York served as precursors to the Great Depression, while local peaks in Illinois coincided with both Stagflation and the Global Financial Crisis. Similarly, in China, Shanghai's skyscraper milestones displayed pronounced sensitivity to the Asian Financial Crisis, with height records closely mirroring the timing of the downturn.

This pattern lends partial support to aspects of the Skyscraper Curse. However, the correlation between skyscraper construction and economic decline is far from uniform. In many cases, towers may instead embody state-led stimulus (Olds, 2014), speculative delays or deliberate strategies of rebranding during times of crisis (Kaika, 2010). Our visualisations indicate that emerging economies such as China and the United Arab Emirates – where development trajectories are more heavily shaped by state intervention – tend to sustain the momentum of skyscraper construction irrespective of broader economic turbulence. By contrast, nations such as the United States, Canada and Germany – where development is more market-driven – exhibit stronger sensitivity to global economic cycles.

A similar dynamic can be observed within nations. Regions that are primarily market-oriented display greater volatility, whereas those supported by central government intervention maintain a steadier trajectory. Shanghai provides a salient example: once a frontrunner, it has now been overtaken by Guangdong, which has pursued a more consistent pace of skyscraper development. This highlights that skyscraper construction is driven by a complex interplay of mechanisms. Any effort to identify universal laws must therefore account for issues of scale, as demonstrated here, which merit further exploration in future research.

Taken together, these findings suggest that skyscrapers are more appropriately interpreted as temporal markers of ambition and intervention rather than deterministic predictors of economic collapse, even though a lagged co-presence between record-breaking heights and economic downturns can often be observed. The extent to which skyscraper construction can be deployed as an indicator of urban economic performance is thus closely tied to prevailing growth modes – modes that not only differ across nations and regions but also evolve over time.

The 3-D urbanisation

Our second visualisation (Figure 2) presents the relative 3-D index for nations and regions, illustrated through diagrams that capture the interrelationship between vertical growth – represented by the cumulative heights of skyscrapers – and horizontal growth – represented by the cumulative urban areas in which these skyscrapers have been built. Among the top 54 nations, the index varies from 0.5 to 1.96, highlighting marked disparities in three-dimensional urbanisation, with variation even more pronounced at the sub-national level. Notably, only Singapore, Bahrain and five other countries attain index values above 1, while most nations cluster around an average of 0.7. Despite belonging to the 'thousand club' of skyscrapers and possessing vast urban areas, China and the United States register comparatively modest scores of 0.89 and 0.85, ranking 13th and 16th, respectively. In practical terms, this means that China requires 167 km² of urban land to produce a single skyscraper of 100 metres, while the United States requires 201 km². Nations with more constrained geographical landscapes or those already exhibiting high levels of urbanisation tend to show higher degrees of three-dimensional growth. This suggests that limitations on horizontal expansion – whether imposed by natural geography or by extensive prior land consumption – constitute a key driver of vertical urbanisation.

Competition for the skyline is more intense among cities than between nations. Nearly 28 subnational regions achieve skyscraper densities exceeding one per 100km², with the lowest index among the top 54 areas still surpassing 0.7. Cities such as Tel Aviv, Ad-Dawah and Kuala Lumpur exceed New York in skyscraper density, as measured by the proposed index (m/km²). The top-ranking list also includes 14 Chinese provinces or metropolitan areas – led by Hong Kong in fifth place – together with 19 US states. Capitals from other nations, such as Moscow, Bangkok, Dubai and São Paulo, also feature prominently. By contrast, European regions remain uniformly low, reflecting a ‘marathoner’ mode of growth. These findings suggest two distinct trajectories: constraint-driven verticalisation in city-states and smaller nations, where land scarcity compels intensive high-rise development; and expansion-driven horizontalism in larger states, where verticality is diluted by an urban economy sustained in part through extensive suburban sprawl.

The construction of skyscrapers thus reflects a form of vertical urbanisation that is not universally necessary but strategically chosen. Such strategies frequently serve political, economic or symbolic purposes and may diverge from the broader economic standing of the country or city in question, illustrating the varied and often context-specific motivations underpinning vertical growth.

Conclusion and discussion

Skyscrapers illuminate the intersections of economy, governance and urban form. They rise not only from demand but as icons of power, capital and global ambition. By visualising global patterns and proposing a relative 3-D urbanisation index, this study demonstrates how vertical and horizontal growth can be systematically compared. Far from curiosities of height, skyscrapers reveal broader urban trajectories in an age of competition and crisis. Integrating vertical and horizontal dimensions provides new insights into how cities expand, compete and reinvent themselves.

Our findings directly address the three research questions. Skyscraper construction mirrors urbanisation but diverges regionally: bursts, prestige peaks and shifting rankings highlight symbolic competition and governance contrasts. Economic correlations exist but are mediated by politics, lags and strategy. The 3-D index offers a novel comparative tool, revealing how vertical and horizontal trajectories intersect unevenly across scales.

The study extends scholarship on vertical urbanism (Appert & Montes, 2015; Harris, 2015), grounding it in comparative visualisation. It reinforces the politics of verticality (Charney, 2007; Graham & Hewitt, 2012), showing that authoritarian regimes and speculative markets yield distinct outcomes. It also supports analyses of skyscrapers as icons of global capitalism (McNeill, 2005; Sklair, 2017; Sudjic, 2005), visible in Middle Eastern ‘Bloomers’ and Asian hubs where towers serve visibility as much as utility.

Our results complement broader morphological analysis (Abel, 2003; Cuthbert, 2011). Skyscrapers show that density is not merely a physical measure but also symbolic, strategic and highly uneven. They embody selective investment, often concentrated in financial or political centres rather than dispersed evenly across metropolitan space.

Policymakers should avoid treating skyscrapers solely as economic barometers. Towers are strategic urban interventions, shaping identity, branding and spatial organisation. The 3-D index provides a heuristic for benchmarking: constrained cities can track how vertical intensification substitutes for horizontal growth, while sprawling megacities can assess whether towers reflect genuine demand or speculative ambition. Sustainability concerns are paramount: skyscrapers reduce land use but heighten energy demand, affordability pressures and socio-spatial segregation.

While comprehensive, the dataset is not without limitations. The 100 m threshold excludes mid-rise buildings, which shape density but are not classed as skyscrapers. Data quality might also vary regionally: North America and East Asia are well documented, but Africa and South America remain sparse. Completion dates reflect finalisation rather than investment decisions, creating lags between planning, economic cycles



Figure 2. Varying interrelationships between the cumulative heights of skyscrapers and the size of urban areas across the top 54 nations and 54 regions, ranked by the relative 3-D urbanisation index. This index is calculated as the ratio of the log-normalised total skyscraper heights to the log-normalised size of the built-up areas

and realised projects. The log-based formulation of the index moderates extreme values but cannot fully capture variations in massing or design. Nevertheless, the dataset remains the most robust comparative source currently available, offering new opportunities for visualising and theorising vertical urbanisation. Future research should test skyscraper–economy relationships statistically, expand coverage to under-documented regions and integrate sustainability and inequality dimensions.

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